Method for Refiguring Pump Performance from Clearance Change

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1. Scope and field of application

This document defines a method to estimate efficiency change due to running clearances modification, reduction or enlargement, of a centrifugal pump.

This procedure is applicable to single- and multi-stage pumps because it is driven by pump specific speed.

The procedure applies to the Nova Milanese and La Spezia plants.

2. Definitions

BEP best efficiency point

N_s specific speed

3. General Procedure - Description

Using Mod.467[1] it is possible to calculate a preliminary estimation of pump efficiency modification as a function of pump specific speed (N_s) when for a certain application it is necessary to:

- enlarge internal running clearances, for example in hot service and correspondingly take into account for worse pump perfomance
- reduce internal clearances with respect to Table 6 of API 610 std.[2] to improve pump performance for example in terms of efficiency or rotordynamic

The method is based on the graph of from which it is possible to obtain a correlation for all the possible combinations of changes listed here:

- 0.75 x API standard clearances
- API 610 clearances
- API 610 hot clearances
- PEEK reduced clearances (60-50% API clearances)

For this last option, PEEK is representative of non-metallic materials for wear rings necessary when clearances are reduced quite a lot in order to avoid galling problems.

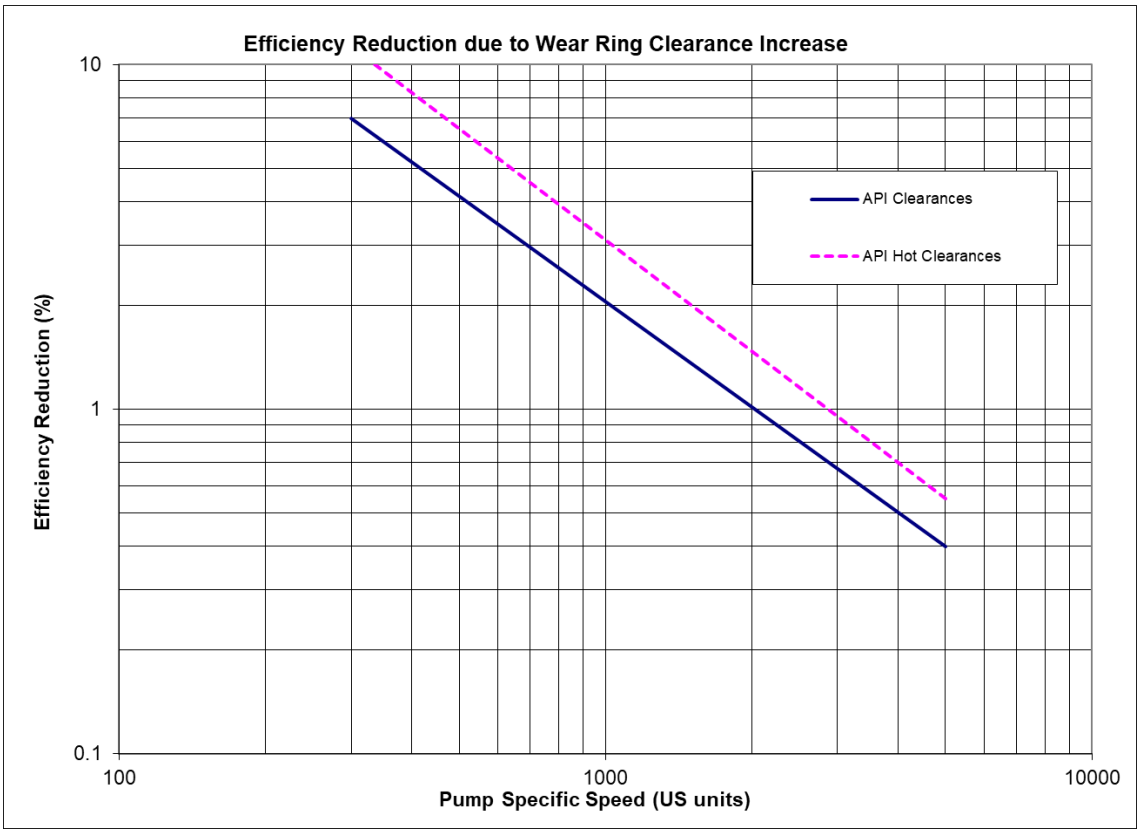


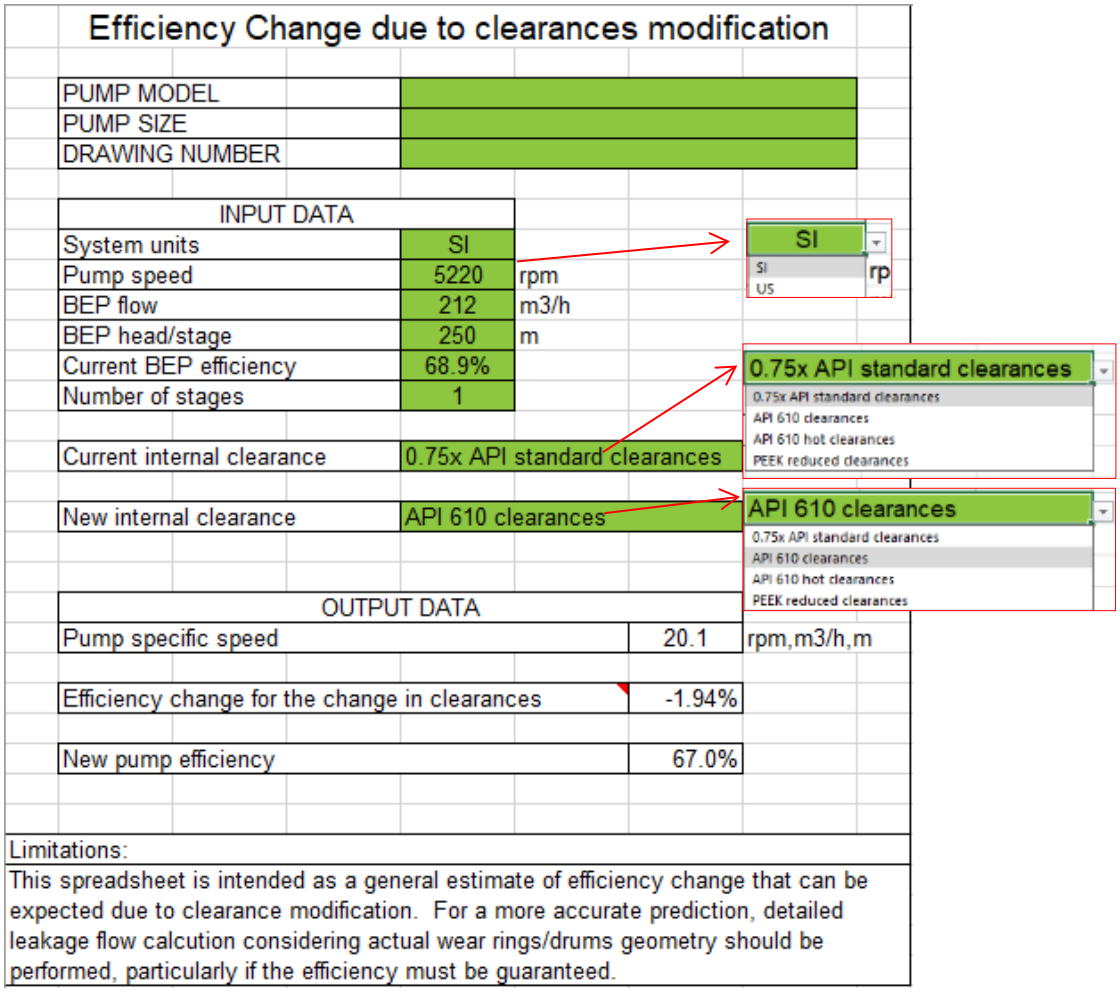
Figure 1. Efficiency Modification due to Clearances Change as a Function of Pump Specific Speed[3]

4. Methodology, analysis and interpretation of results

Only few input data (green cells of Figure 2) are necessary for the calculation spreadsheet

- System units: it is possible to use SI or US units
- Pump speed, rpm
- BEP flow, m^3/h or USgpm
- BEP head, m or ft
- Current BEP efficiency
- Number of stages

Then the user shall select current internal clearances used in the pump with the data provided and the desired new configuration of internal clearances choosing from the four available options (see Figure 2).



Efficiency Change due to clearances modification

PUMP MODEL	
PUMP SIZE	
DRAWING NUMBER	

INPUT DATA		
System units	SI	
Pump speed	5220	rpm
BEP flow	212	m3/h
BEP head/stage	250	m
Current BEP efficiency	68.9%	
Number of stages	1	
Current internal clearance	0.75x API standard clearances	
New internal clearance	API 610 clearances	

OUTPUT DATA		
Pump specific speed	20.1	rpm,m3/h,m
Efficiency change for the change in clearances	-1.94%	
New pump efficiency	67.0%	

Limitations:
This spreadsheet is intended as a general estimate of efficiency change that can be expected due to clearance modification. For a more accurate prediction, detailed leakage flow calcution considering actual wear rings/drums geometry should be performed, particularly if the efficiency must be guaranteed.

Figure 2: Input/Output Data of Mod.467 for Efficiency Change Calculation due to Clearance Change

The output section shows calculated pump Ns, the increase or decrease of efficiency and the new pump efficiency at BEP estimated achievable with the change in clearances. The other points of the efficiency curve can be adjusted proportionally.

5. Acceptance Criteria

This method is useful for a preliminary estimation of efficiency change due to change in internal running clearances with these four options:

- 0.75 x API standard clearances
- API 610 clearances

- API 610 hot clearances
- PEEK reduced clearances (60-50% API clearances)

The method has been tested with few experimental and CFD data available and results give good agreement with test data. However, for a more accurate prediction, detailed leakage flow calculation considering actual wear rings/drums geometry should be performed, particularly if the efficiency must be guaranteed.

6. References

[1] Mod.467 – Method for Refiguring Pump Performance from clearance change

[2] *API STANDARD 610, 12th Edition, January 2021*

[3] ANSI/HI 1.3-2000, Figure 1.78B – Estimated efficiency decrease due to increased wear ring clearance